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In cats, males are XY and females are XX. A gene on the X chromosome controls fur colour in cats. The allele **G** codes for ginger fur and the allele **B** codes for black fur. These alleles are codominant. Heterozygous females have ginger and black patches of fur and their phenotype is described as tortoiseshell female.

The two alleles, **F** and **f** of a different gene, which is **not** sex-linked, interact with the gene controlling fur colour. The allele **F** is dominant and stops the formation of pigment in the fur, resulting in white fur. The allele **f** is recessive and has no effect on fur colour.

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1

Name the type of interaction between the two genes affecting fur colour.

[1 mark]

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2

What phenotype would a cat with the following genotype have?

[1 mark]

$X^G X^B ff$ _____

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3

Complete the genetic diagram to show all the possible genotypes and the ratio of phenotypes expected in the offspring of this cross.

[3 marks]

Phenotypes of parents

$X^G X^G Ff$

x

$X^B Y ff$

Genotypes of offspring _____

Phenotypes of offspring _____

Ratio of offspring phenotypes _____

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In a population, 36% of cats had the **F** allele and had white fur.

Use the Hardy–Weinberg equation to calculate the frequency of the **f** allele in this population.

Show your working.

[2 marks]

Answer _____

Turn over for the next question

